

- Q.28 Define "Absolute Humidity" and "Relative Humidity."
- Q.29 Describe the industrial weighing system using Strain Gauges.
- Q.30 How is conductivity used to check water purity?
- Q.31 Differentiate between Open-Loop and Closed-Loop control.
- Q.32 Explain the Proportional Band.
- Q.33 Discuss the limitations of On-Off control.
- Q.34 What is the working principle of a pH meter?
- Q.35 Define "Instrument Lag" and "Fidelity".

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the construction, principle, and working of a Resistance Temperature Detector (RTD).
- Q.37 Describe the Bourdon tube pressure gauge in detail with its advantages and disadvantages .
- Q.38 Explain the concept of Controller Tuning. Describe the Ziegler-Nichols method in detail.

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5th Sem / Food Technology Sub : Instrumentation and Process Control

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The difference between upper and lower scale limits is:
a) Range b) Span
c) Fidelity d) Drift
- Q.2 RTD usually uses which metal?
a) Copper b) Iron
c) Platinum d) Alumel
- Q.3 Which gauge is used for measuring vacuum?
a) McLeod Gauge b) Bourdon Gauge
c) Diaphragm d) U-Tube
- Q.4 Manometers use the principle of:
a) Hydrostatic balance b) Thermal expansion
c) Electromagnetism d) Vibration
- Q.5 In an Orifice meter, the point of minimum area is called:
a) Stagnation point b) Vena Contracta
c) Throat d) Discharge point

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- Q.6 A Pitot tube measures:
 a) Static pressure b) Impact pressure
 c) Both 'a' & 'b' d) Temperature
- Q.7 Conductivity of a liquid depends on:
 a) Ion concentration b) Colour
 c) Volume d) Smell
- Q.8 Hygrometers measure:
 a) Flow b) Pressure
 c) Humidity d) pH
- Q.9 A controller that maintains output between two limits is:
 a) Proportional b) On-Off
 c) Integral d) Derivative
- Q.10 Automatic process control aims to keep the Process Variable at the:
 a) Dead band b) Set point
 c) Zero d) Maximum

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ is the closeness of repeated measurements to each other.
- Q.12 A thermistor is made of _____ material.
- Q.13 McLeod gauge is used for measuring _____.

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- Q.14 The ratio of actual flow to theoretical flow is the Coefficient of _____.
- Q.15 P-Control always results in a steady-State _____.
- Q.16 Thermocouples are active transducers (Generate their own voltage). (True / False)
- Q.17 A venturi meter is longer in size than an Orifice meter. (True / False)
- Q.18 Conductivity measurement requires two electrodes. (True / False)
- Q.19 Derivative control can be used alone for any process. (True / False)
- Q.20 Dynamic error occurs when the input is changing rapidly. (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define Sensitivity and explain its significance.
- Q.22 List three materials used for thermocouples and their ranges.
- Q.23 Explain the working of a thermistor.
- Q.24 Describe a U-Tube Manometer with a diagram.
- Q.25 Explain the "Different Method" for liquid level measurement.
- Q.26 What are the advantages of a Venturi meter?
- Q.27 Explain how a Pitot tube is used in a pipe flow.

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